

Problem Set

1. Create a class Book with two instance variables: title and author. Create an object of the Book class in the main method, set the values for the variables, and print the title and author.
2. Write a class Student with two instance variables: name and rollNo. Create a constructor to initialize these values. Create an object and print the student details using a function named printDetails().
3. Create a class Rectangle with two attributes: length and width. Define a method getArea() to return the area of the rectangle. Create an object of the class, set the values, and call the method to print the area.
4. Define a class Employee with a static variable companyName and an instance variable employeeName. Use a static method to set the company name and a non-static method to display employee details.
5. Create a class Account with private attributes balance and accountNumber. Provide public getBalance() and setBalance() methods to access and modify the balance. Create an object and test the functionality.
6. Create a class Person with an attribute name and a method display(). Then create a subclass Student that adds an attribute studentId. Override the display() method to print both name and studentId.
7. Define a class Calculator with multiple add() methods that take different numbers or types of parameters (int, double, etc.). Implement method overloading by adding methods that return the sum of two numbers.
8. Create a class MathUtils with a static method max() that returns the maximum of two integers. Call this static method from the main() method without creating an object of MathUtils.
9. Create a class Product with attributes name and price. In the main() method, create an array of 5 Product objects, initialize them with values, and print the product names and prices.
10. Write a class Complex to represent complex numbers. Implement a method add(Complex c) that adds another complex number to the current one. Create two objects, add them, and display the result.
11. Create a class Vehicle with overloaded constructors: one with no parameters, one with a String model, and one with both String model and int year. Demonstrate the use of all three constructors in the main() method.
12. Define an abstract class Shape with an abstract method area(). Create two subclasses, Circle and Rectangle, and override the area() method in each class to calculate the area based on the shape's attributes.

13. Create an interface Playable with a method play(). Implement this interface in two classes: Guitar and Piano. In the main() method, demonstrate polymorphism by calling the play() method on objects of both classes.
14. Create two classes: Department and Student. The Department class contains a list of students. Write a method addStudent(Student s) in the Department to add a student to the department. Demonstrate aggregation by adding students to a department and displaying the students.
15. Create two interfaces Teacher and Researcher, both containing a method work(). Create a class Professor that implements both interfaces and provides its own implementation of work(). Call the work() method from a Professor object.